1.

(1) Each map $\Delta^0 \to X$ corresponds to a point of X, so $C_0(X)$ is free on the set X. $C_1(X)$ is generated by continuous maps $[0,1] \to X$, i.e. paths in X, and the image of a path $f : [0,1] \to X$ under ∂ is $\partial f = f(1) - f(0)$. Hence $H_0(X) = C_0(X)/\mathrm{Im}\partial$ is the free group on the components of X, since every two point in the same component is connected by a path.
(2) Analogously, consider $\iota : C_0(X) \to \mathbb{Z}$ by taking each generator to 1, then $\tilde{H}_0 = \mathrm{Ker}\iota/\mathrm{Im}\partial$ where $\mathrm{Ker}\iota$ is generated by elements $x - x_{\lambda_0}$ for $x \in X$, and two elements $x - x_{\lambda_0}$ and $y - x_{\lambda_0}$ have the same image in $\tilde{H}_0$ iff x, y are in the same component. Therefore $\tilde{H}_0(X)$ is free on $x_\lambda - x_{\lambda_0}$ where $\lambda \neq \lambda_0$.

2.

Let $Z_r(K) = \ker(\partial : K_r \to K_{r-1})$. Apply the snake lemma to

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A_q & \xrightarrow{i} & B_q & \xrightarrow{p} & C_q & \longrightarrow & 0 \\ & & \downarrow \partial & & \downarrow \partial & & \downarrow \partial & & \\ 0 & \longrightarrow & Z_{q-1}(A) & \xrightarrow{i} & Z_{q-1}(B) & \xrightarrow{p} & Z_{q-1}(C). & & \end{array}$$

The lower row is exact; surjectivity at its right end is not required. The vertical kernels are $Z_q(K)$, and their cokernels are $H_{q-1}(K)$. Thus we obtain the exact sequence

$$Z_q(B) \longrightarrow Z_q(C) \xrightarrow{\hat{\delta}_q} H_{q-1}(A) \xrightarrow{i_*} H_{q-1}(B) \xrightarrow{p_*} H_{q-1}(C).$$

Explicitly, for $c \in Z_q(C)$, choose $b \in B_q$ with $p(b) = c$. Exactness gives a unique $a \in A_{q-1}$ with $i(a) = \partial b$; injectivity of i gives

$$i(\partial a) = \partial^2 b = 0, \qquad \text{so } \partial a = 0.$$

The snake map is $\hat{\delta}_q(c) = [a]$, independent of the lift and additive.
If $c = \partial c'$, lift c' to b' and choose $b = \partial b'$; then $a = 0$. Hence $\hat{\delta}_q$ kills boundaries and descends to $\delta_q : H_q(C) \to H_{q-1}(A)$.
Quotienting the cycle terms by boundaries preserves exactness at $H_q(C)$. Exactness at $H_{q-1}(A)$ and $H_{q-1}(B)$ already follows from the snake lemma. Varying q yields

$$\cdots \longrightarrow H_q(A) \xrightarrow{i_*} H_q(B) \xrightarrow{p_*} H_q(C) \xrightarrow{\delta_q} H_{q-1}(A) \longrightarrow \cdots.$$

3.

By Mayers-Vietoris we have $\tilde{H}_1(\mathbb{S}^n) \to \tilde{H}_0(A \cap B) \to \tilde{H}_0(A) \oplus \tilde{H}_0(B)$ where the two ends are zero since A, B are connected open subsets of a manifold. Hence $\tilde{H}_0(A \cap B) = 0$ and $A \cap B$ is connected.

4.

Note that $\mathbb{S}^n \setminus \mathrm{pt} = \mathbb{R}^n$ is contractible, and $\mathbb{R}^n \setminus \{p_1, \cdots, p_k\}$ deformation retracts to $\bigvee_{j=1}^k \mathbb{S}^{n-1}$. Hence the nonzero homology groups of $X = \mathbb{S}^n \setminus \{p_1, \cdots, p_{k+1}\}$ are $H_{n-1}(X) = \mathbb{Z}^k$ and $H_0(X) = \mathbb{Z}$. (In the case $n = 1$, only $H_0(X) = \mathbb{Z}^{k+1}$.)

5.

(a) Let U, V be small neighborhoods of the two $\mathbb{S}^n$ around the equator, that deformation retracts to $\mathbb{S}^n$, and their intersection deformation retracts to $\mathbb{S}^{n-1}$. Then MV sequence gives

$$\cdots \to H_k(\mathbb{S}^{n-1}) \to H_k(\mathbb{S}^n) \oplus H_k(\mathbb{S}^n) \to H_k(X) \to \cdots$$

(b) The only non-zero terms of reduced homology gives $0 \to \mathbb{Z}^2 \to \tilde{H}_n(X) \to \mathbb{Z} \to 0$. In fact this short exact sequence splits since $\mathbb{Z}$ is a projective module. Therefore $\tilde{H}_n(X) = \mathbb{Z}^3$, so the only non-zero homology groups are $H_n(X) = \mathbb{Z}^3$ and $H_0(X) = \mathbb{Z}$.

6.

To construct a map $f : \mathbb{S}^n \to \mathbb{S}^n$ of any given degree d, first note that $z \mapsto z^d$ gives a map of degree d on $\mathbb{S}^1$, so it suffices to show that any $f : \mathbb{S}^n \to \mathbb{S}^n$ and $Sf : \mathbb{S}^{n+1} \to \mathbb{S}^{n+1}$ have the same degree, by identifying $\mathbb{S}^{n+1}$ as $S\mathbb{S}^n$, the quotient of $\mathbb{S}^n \times [-1, 1]$.
Consider $f : X \to X$ and $Sf : SX \to SX$, then SX is the union of two hemispheres U, V, which are contractible cones and $U \cap V = \mathbb{S}^n$. Hence for $X = \mathbb{S}^n$, MV sequence gives the natural isomorphism $\tilde{H}_{n+1}(\mathbb{S}^{n+1}) \to \tilde{H}_n(\mathbb{S}^n)$. Note that S is natural, i.e. Sf, f commutes with $X \times [-1, 1] \to SX$. Hence by naturality $(Sf)_*, f_*$ commutes with $\tilde{H}_{n+1}(\mathbb{S}^{n+1}) \to \tilde{H}_n(\mathbb{S})^n$, i.e. Sf has the same degree with f.

7.

(a) Recall that the topologists sine curve is connected but not path-connected, so we need only consider the two components U, V, which are both contractible. Hence the only non-zero homology group is $H_0(X) = \mathbb{Z}^2$.
(b) Cut the curve at point a: let $U_a = \{x \in T : x_1 \leqslant a\}$ and $V_a = \{x \in T : x_1 \geqslant a\}$, then as subsets of $\mathbb{S}^n$, $U_a^C \cup V_a^C = \mathrm{pt}^C = \mathbb{R}^n$ and $U_a^C \cap V_a^C = T^C$. Hence MV sequence gives $H_q(\mathbb{S}^n \setminus T) \to H_q(U_a^C)$ $\oplus H_q(V_a^C) \to H_q(\mathbb{R}^n)$, so $H_q(\mathbb{S}^n \setminus T) = H_q(\mathbb{S}^n \setminus V_a)$. (The complement of a line segment has zero homology, see Hatcher 2B.1). As $a \to 0$, V_a shrinks to the vertical line segment, so since V_a has compact image, $H_q(\mathbb{S}^n \setminus T)$ is the same as $H_q(\mathbb{S}^n \setminus V_0) = 0$.

8.

Suppose there is a nowhere vanishing vector field v on $\mathbb{S}^{2n}$, then consider the antipodal map $f = -1$ on $\mathbb{S}^{2n}$ which has degree $(-1)^{2n+1} = -1$. v gives a homotopy $H_t(x) = x \cos t + \frac{v(x)}{\|v(x)\|} \sin t$ on $\mathbb{S}^{2n}$, between 1 and -1, but they have different degree, a contradiction.